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RESEARCH

Built-up area mapping for sustainable urban planning: a novel satellite imagery based approach for detection of urbanization and landscape changes

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Abstract

Accurate estimation of built-up areas provides valuable information on urbanization trends and infrastructure development, which enables policymakers to formulate decisions in the planning and management of urban infrastructure, resource allocation, and environmental management. Although many remote sensing-based indices are widely used to map built-up areas and monitor the impact of urbanization on climate change, the close resemblances in the spectral features of built-up areas with bare soil areas pose a great challenge in the precise delineation of built-up areas, which may result in their overestimation. Moreover, most of the existing built-up indices use thresholding techniques to discriminate built-up areas from bare soil, but their effectiveness is limited due to the spatio-temporal variability that affects mapping accuracy. The present research introduces a novel constraint-based built-up indexing (AMCBI) method that improves the distinction between target and background features with similar spectral signatures, eliminating the need for thresholding, and demonstrates its potential over existing spectral indices for precise mapping applications. The qualitative and quantitative analysis of AMCBI revealed overall agreement values of above 97% and an F1 score above 0.95 across all study sites. Further, the weak correlation between AMCBI and existing indices indicates reduced spectral confusion and eliminates the need for thresholding or masking techniques. The superiority of the AMCBI can be mainly attributed to the ability of the constraint-based indexing technique to de-correlate built-up areas from bare soil. The findings of the study suggest that AMCBI is a viable alternative in mapping built-up areas, especially in heterogeneous urban landscapes. Clinical trial number: not applicable.

Keywords Built-up area · Bare soil · Mapping · Index · Remote sensing · Threshold

Introduction

The unprecedented increase in industrialization and urbanization greatly impacts the spatial dynamics of urban sprawl. Though the rapid urban agglomeration has resulted in economic development and human well-being, it has led to environmental and ecological concerns, including natural habitat destruction, loss of biodiversity, decreased drainage capacity, and water quality degradation, which require further investigation (Pringle 2001). Moreover, the effect of urban heat islands aggravates extreme weather events, imposing more strain on resources and infrastructure. In this context, urban area mapping enables policymakers to monitor urban sprawl and land cover changes, identify growth patterns, and help in disaster management and environmental monitoring for a resilient and sustainable future.

Conventional mapping methods, such as ground surveys and aerial photography, require many resources (Richards 2013), which may not be effective in regions of high urban expansion (Small 2003; Perepechko et al. 2005; Lein 2006). Remote sensing data has been mainly employed for mapping

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different types of land cover using pixel and object-based classification methods and indices-based methods (Zha et al. 2003; Knight et al. 2006). The global level implementation of object-based classification poses challenges related to the complexity of optimizing segmentation parameters, whereas pixel-based classification with the mixed pixel problem leads to significant errors (Lu et al. 2013; Yu et al. 2017). The implementation of spectral mixture analysis (SMA) based methods is complex due to the difficulties involved in accurately selecting end members and quantifying intra-class variability (Lu and Weng 2004; Powell et al. 2007). However, the indices-based method, which involves direct segmentation of the imagery, has been widely employed over other methods due to the ease of implementation in terms of time and effort (Ali and Nayyar 2021). The indices are formulated by analyzing the spectral signature of the target area from satellite images to delineate the feature of interest (Huete and Jackson 1987; Dozier 1989). Although numerous spectral indices have been formulated and reported in the literature (Kawamura et al. 1996; Zha et al. 2003; Deng and Wu 2012; Bhatti and Tripathi 2014; Liu et al. 2014; Estoque and Murayama 2015; Luo et al. 2015; Sun et al. 2016; Bouhennache et al. 2019), to enhance the efficiency in built-up surface delineation from remote sensing data, the differentiation between built-up and bare land is difficult due to their spectral similarity, often leading to frequent misclassification (Piyoosh and Ghosh 2018).

Built-up Area Extraction Index (BAEI) (Bouzekri et al. 2015), introduced in the recent past, employs the green, red, and Short-Wave Infrared (SWIR1) bands combined with a constant (L) to extract urban built-up features effectively. However, the result of the BAEI analysis depicted confusion between built-up features and rocks (Pandey and Tiwari 2020). Built-up Surface Index (BSI) (Sameen and Pradhan 2016) employed yellow and Near-Infrared (NIR) bands of WorldView-2 imagery to delineate built-up surfaces with the Particle Swarm Optimization (PSO) technique for the selection of relevant bands. The multi-index approach (Ettehadi Osgouei et al. 2019) employed by combining the Modified Normalized Difference Water Index (MNDWI), Normalized Difference Tillage Index (NDTI), and Normalized Difference Vegetation Index (NDVI) has been employed to distinguish urban surfaces from bare lands using Sentinel-2A imagery effectively. Automated Built-up Extraction Index (ABEI) has been developed to enhance the delineation of built-up regions with bare and sandy terrains using an improved gravitational search algorithm (IGSA) optimization to fix the suitable weights for each of the Landsat 8 OLI imager bands (Firozjaei et al. 2019). Meanwhile, the Modified Built-up Index (MBI) was formulated by integrating the Soil Adjusted Vegetation Index (SAVI), Normalized Difference Water Index (NDWI), Advanced Slope-Based Indexing Technique Built-Up Index (ASIT_BI), and Brightness Temperature (TB) to enhance the delineation of built-up features using Landsat 8 Imagery (Ali and Nayyar 2021).

The current advancements in built-up area mapping techniques are largely limited to manual or semi-automated procedures that depend on optimal thresholding-based methods. Moreover, built-up area mapping is challenging due to the complexity of urban landscapes and overlapping spectral characteristics with bare soil (Gamba et al., 2003; Herold et al. 2004; Lu and Weng 2004; Powell et al. 2007; Lwin and Murayama 2013). The existing built-up indices require precise masking of spectrally similar bare soil areas and optimal threshold selection for accurate delineation of built-up areas. However, this requirement poses a great challenge due to the high variability in spatial, temporal, and spectral characteristics of urban areas, limiting the effectiveness of these indices in mapping built-up regions. To address this, it is vital to analyze the spectral response of urban areas across multiple bands and sensors (Maynard et al. 2007) and utilize high-resolution data (Thomas et al. 2003; Lwin and Murayama 2013). These limitations have prompted researchers to explore alternative approaches, including the combination of multiple indices to enhance built-up area extraction (Bhatti et al. 2014). However, combining indices and thresholding methods are insufficient to separate built-up areas from non-built-up areas due to heterogeneous spectral signatures, spectral variability (Kaimaris et al. 2016), and difficulties distinguishing from bare soil, rock surfaces, sandy soils, and bare land (As-syakur et al. 2012; Bhatti et al. 2014; Firozjaei et al. 2019). The urban heat island effect, which increases land surface temperatures in urban areas, can be a valuable indicator of urban intensity (Oke 1982; Lo 2004; Weng et al. 2004; Yuan and Bauer 2007), but the accuracy of mapping urban areas using thermal data is restricted by its limited spatial resolution, particularly for larger areas.

Although the existing built-up indices formulated using discrete bands of multispectral data are effective to some extent, the ambiguity between barren land and the built-up area has only been partially addressed (Pandey and Tiwari 2020). Hence, a new constraint-based indexing technique without the need for thresholding is proposed in this study to enhance the delineation of target features from the background, having similar spectral patterns, with improved accuracy in mapping built-up areas. The developed methodology, AMCBI, is tested and validated in six different environmental backgrounds and climatic conditions of the Indian subcontinent using Sentinel 2 images. The performance of the new methodology is assessed both qualitatively and quantitatively by comparing it with the widely used built-up indices such as BAEI, NBI, and NDBI. Further, the seasonal variability and the environmental application of the newly developed indexing technique are also evaluated.

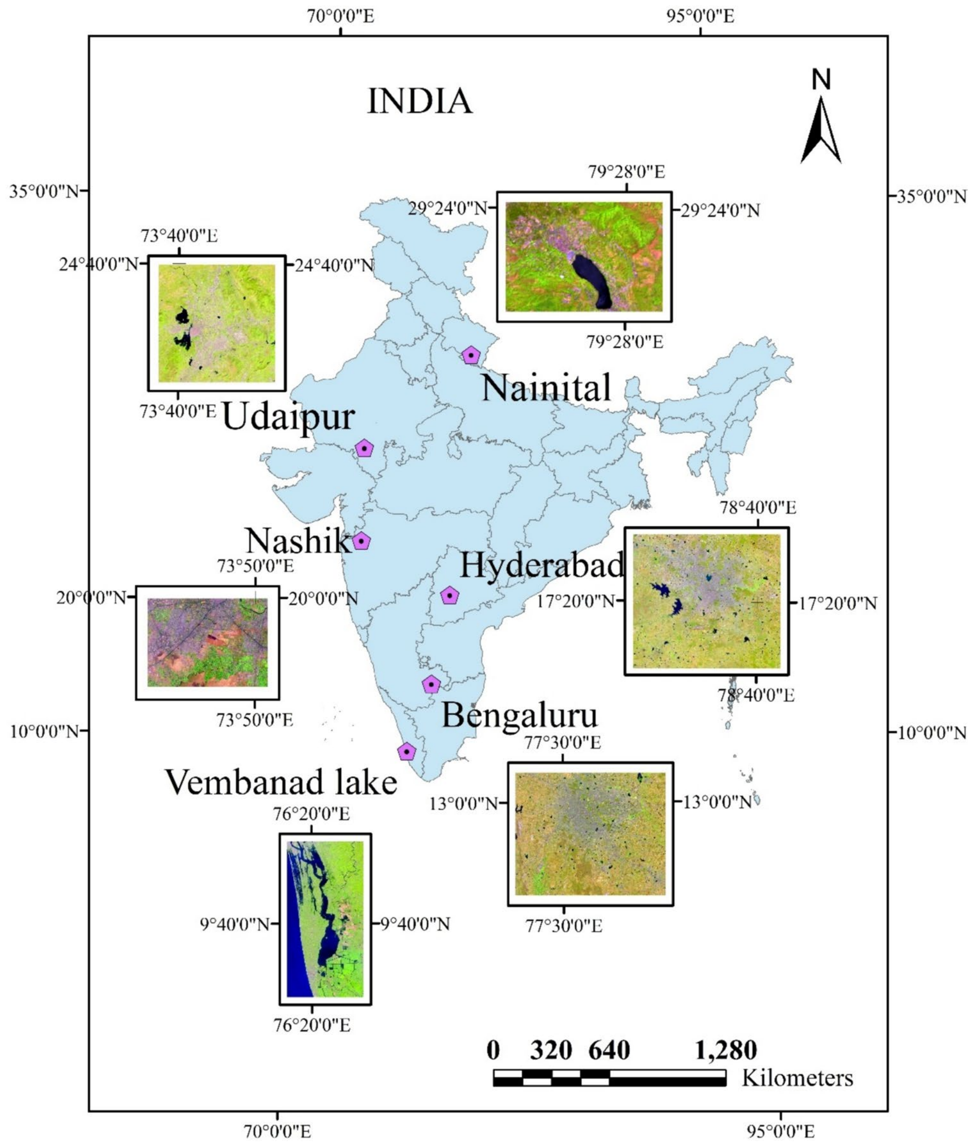

Fig. 1 Location of the chosen study areas

Study area and data

The effectiveness of the proposed methodology (AMCBI) is assessed in two test sites and four validation sites. These study areas (Fig. 1) are strategically selected to represent diverse environmental conditions and climate zones across the Indian subcontinent, with a focus on regions experiencing rapid urbanization. An overview of the locations and details of all six study sites is provided in Fig. 1 and Table 1. The region around Vembanad Lake in Kerala, India, has been chosen as one of the test sites due to its unique setting of urban built-up areas, bare land, and vegetative zones near the coastal area of central Kerala, which is ideal for the study. Another test site, Nainital, located at an elevation of around 2000 m, is situated in the Lesser Himalaya of Uttarakhand State and is surrounded by peaks such as Naina Peak, Ayarpatha, and Deopatha. It has a subtropical climate and winter temperatures that can drop as low as 0.9 °C in the hilly regions. The four validation sites chosen (Table 1)

Table 1 Description of four study sites

Study area	Climate type	Imagery date	Degree of built-up area	Degree of bare soil	Degree of vegetation
Vembanad lake	Tropical monsoon	07/02/2021	Low	Medium	High
Nainital	Sub-tropical humid	26/04/2021	Medium	Low	High
Nasik	Sub-tropical humid	13/02/2024	High	Low	Medium
Udaipur	Tropical arid	14/10/2021	High	Medium	Low
Bengaluru	Sub-tropical steppe	23/01/2024	High	Low	Low
Hyderabad	Sub-tropical steppe	04/12/2021	High	Low	Low

are located in regions experiencing rapid urban expansion amidst diverse environmental conditions, making them ideal for in-depth analysis. The study employs recent cloud-free Sentinel-2 imagery (Table 1), processed to Level 2A, from the Copernicus hub (https://www.copernicus.eu/en/access-data/conventional-data-access-hubs). These images have undergone geometric, radiometric, and atmospheric corrections to ensure a maximum number of valid pixels for accurate analysis. For the quantitative assessment of built-up areas, reference data is generated by the integration of field observations with manual digitization of high-resolution images of Google Earth Engine to provide a robust and accurate ground truth dataset. To ensure accuracy, the precisely classified built-up areas are cross-validated with the results from the maximum likelihood classification method, thereby minimizing potential errors in the reference dataset.

Methodology

This study formulated a new constraint-based built-up area mapping technique to maximize the elimination of bare land/soil that is spectrally similar. The accurate elimination of bare soil area ensures that the built-up area is not overestimated/underestimated, thereby providing a more precise built-up area mapping.

Spectral characteristics of various land covers

The development of a methodology to differentiate built-up areas from bare land/soil using satellite imagery requires the collection and analysis of representative samples from major types of land cover, which include vegetation, water, built-up areas, bare soil, and rocks. The spectral responses of these samples are compared with their corresponding pixels from satellite imagery, which enables the identification of distinctive characteristics useful for accurate land cover classification. Spectral bands with the highest contrast between built-up areas and surrounding features are identified and selected based on the largest difference in mean reflectance values (Fig. 2 (a)). Built-up areas consist of buildings, roads, parking lots, and surfaces made of stone, concrete, bricks, asphalt, and cemented surfaces, which represent transformed land surfaces (Liu et al. 2022; Zhao and Zhu 2022; Smitha et al. 2024). Moreover, built-up areas exhibit diverse spectral reflectance patterns due to the complexity and heterogeneity of urban characteristics.

Analysis of spectral responses (Fig. 2 (a)) reveals that built-up areas exhibit maximum reflectance in the SWIR1 band and minimum reflectance in the blue band. Therefore, a combination of SWIR1 and blue bands can effectively enhance the delineation of built-up areas. Nevertheless, bare soil and rocks also exhibited large differences between SWIR1 and blue, which introduce uncertainties and complexities in the accurate separation of built-up areas. Further, a greater enhancement in surface reflectance values from the red to NIR region for bare soil, rocks and vegetated areas (Fig. 2 (a)) is also noted. However, the variability in reflectance among different bands (Fig. 2 (a)) is found to be less for the built-up area. Built-up areas exhibited a unique scatter distribution that is closely aligned with the 1:1 trend line, in the scatter plot between the blue and SWIR1 bands (Fig. 2 (b)). In contrast, bare soil areas are scattered high above the trend line, which occupies a distinct region. Figure 2 (b) effectively illustrates the potential of the combination of blue and SWIR bands for the formulation of appropriate criteria/constraints in the elimination of bare soil areas from built-up areas. Therefore, the integration of SWIR and blue bands, along with red and NIR bands, in the AMCBI formulation enables more effective differentiation and easier delineation of urban areas from other prominent types of land cover.

Development of advanced multispectral constraint-based built-up index (AMCBI)

Existing indices distinguish target areas from background features based on normalized differences between spectral bands with high and low reflectance (Xu 2010). However, they fail to adequately suppress background features with

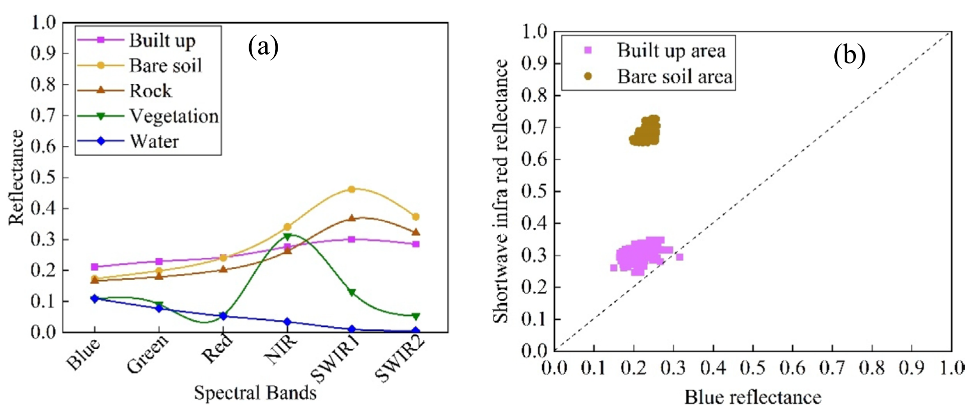

Fig. 2 (**a**) Statistical mean of the reflectance for the prominent types of landcover (**b**) Scatter plot between the reflectance of blue and SWIR for built-up and bare soil area

similar spectral signatures, which results in limited accuracy. Nevertheless, index-based methods currently depend on optimal thresholding to achieve mapping accuracy (Dixit et al. 2019; Luo et al. 2015) but are limited due to their labor-intensive and highly subjective nature (Kaushik et al. 2022). This study proposes a new constraint-based approach, AMCBI, to enhance the differentiation between target and non-target areas with similar spectral signatures without the need for thresholding (Fig. 3). This approach involves a two-step process: (i) spectral transformation, which employs the normalized difference of selected bands, and (ii) constraint-based logical elimination, which removes background features based on the spectral characteristics of the study area. The spectral transformation (Eq. 1) integrates the largest and smallest spectral reflectance bands to maximize the difference between target and non-target features, thereby enhancing their distinguishability. This approach yields positive values for target features and negative values for non-target features. Moreover, a normalized difference index helps emphasize a target feature by maximizing the contrast between the strongest and weakest spectral responses of the feature, reduces uncertainties in surface reflectance estimation, and makes it more reliable than using a simple band ratio. (Du et al., 2015; Zhao et al. 2022). A constraint-based logical elimination technique is then employed to further refine the target feature extraction, using the blue and SWIR1 bands ratio (Eq. (2)) to eliminate background noise and enhance feature delineation. The final values of the AMCBI are determined by applying Eq. (3).

$$AMCBI^{*} = \frac{(SWIR1 + Red) - (NIR + Blue)}{(SWIR1 + Red) + (NIR + Blue)} \tag{1}$$

$$k = \frac{Blue}{SWIR1} \tag{2}$$

$$\begin{gathered}\mathrm{AMCBI} = \mathrm{if}((\mathrm{AMCBI}^{*} > 0)\ \ (0.6 >= \mathrm{k} >= 1)) \\ the\mathrm{AMCBI}^{*} else\,(\_\ |\mathrm{AMCBI}^{*}|)\end{gathered} \tag{3}$$

where ‘*k*’ is the constraint parameter to ensure complete elimination of the separation of built-up areas from bare land features.

The value of ‘*k*’ is fixed based on the spectral analysis of built-up and bare soil areas for blue and SWIR1 bands. In contrast to soils, built-up areas showed higher reflectance in the blue band and lower reflectance in SWIR1, which helps to separate built-up areas from spectrally similar non-built-up areas such as soils (Fig. 2b). The value of ‘k’ less than 0.6

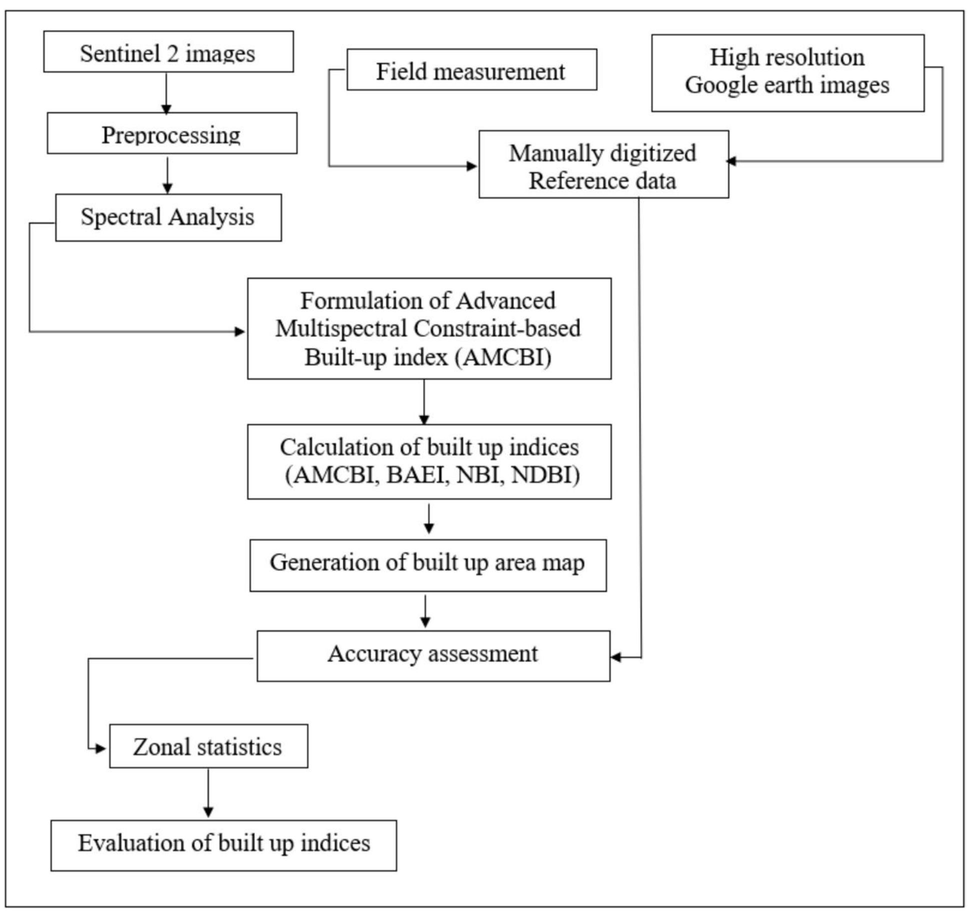

Fig. 3 Methodology adopted in the study

represents non-built-up areas, whereas the ‘k’ value greater than 0.6 represents built-up areas. The AMCBI values range from −1 to 1, with positive values indicating built-up areas and negative values indicating non-built-up areas. To compare the performance of AMCBI, three commonly used built-up indices, namely BAEI, NBI, and NDBI, are used. The Built-up Area Extraction Index (BAEI) is a spectral index that uses green, red, and SWIR1 reflectance, along with a constant ‘L’, to estimate built-up areas (Kumar et al. 2015). The New Built-up Index (NBI) of Jieli et al. (2010) employs spectral signatures from the red, NIR, and SWIR1 bands. The Normalized Difference Built-up Index (NDBI), developed by Zha et al. (2003) to identify built-up regions, employs NIR and SWIR1 bands. The mathematical expressions of BAEI, NBI, and NDBI are given below.

$$BAEI = \frac{Red + L}{Green + SWIR1} \tag{4}$$

$$NBI = \frac{Red\text{x}SWIR}{NIR} \tag{5}$$

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR} \tag{6}$$

Performance assessment is carried out using reference data extracted from high-resolution satellite imagery in Google Earth Engine and field data to evaluate the accuracy of the built-up map generated using different indices. The test samples, distributed evenly across the image, comprise of 495, 220, 207, and 102 points for built-up, bare soil, vegetation, and water, respectively. The performance of AMCBI is quantitatively evaluated using a range of metrics, including spectral separation metrics, namely the Spectral Discrimination Index (SDI), Jeffries-Matusita Distance (JD) values, Bhattacharyya distance (BD), mapping accuracy measures such as overall agreement, quantity disagreement, and allocation disagreement with reference to established methods (Pontius Jr and Millones 2011; Pickard et al. 2017) and F1 scores.

SDI is a metric used to evaluate the spectral separability between target and non-target areas, with values less than 1 indicating poor separation and values greater than 1 indicating effective separation (Kaufman and Remer 1994; Pereira 1999). SDI values are computed using Eq. (7) (Kebede et al. 2022).

$$SDI = \frac{|\mu_1 - \mu_2|}{\sigma_1 + \sigma_2} \tag{7}$$

where μ_1 and μ_2 denote the mean index values for built-up and non-built-up areas, respectively, and σ_1 and σ_2 represent the corresponding standard deviations of the index values for built-up and non-built-up areas.

BD (Tolpekin et al. 2009), which measures the separation between distinct populations, ranges from 0 to infinity, with the value of 0 indicating complete overlap between classes. To enhance BD, the Jeffries-Matusita (JD) distance, which assesses the separability of probability distributions by normalizing the output between 0 and 2, provides a more refined criterion for evaluating classification output (Daboor et al. 2014).

Results

A combination of qualitative and quantitative evaluations is employed to assess the performance of AMCBI. Separability metrics are estimated for AMCBI images at test and validation sites to determine the ability to distinguish built-up areas from other land cover types. Furthermore, a thorough comparative analysis with three other built-up indices is conducted to precisely assess the efficacy and superiority of AMCBI in urban mapping applications.

Visual evaluation of built-up/urban area maps

The built-up area mapping methods are initially compared qualitatively through the visual examination of the built-up area maps (Fig. 4). The built-up area maps derived by AMCBI are assessed against those produced by BAEI, NBI, and NDBI, using false-color Sentinel-2 images as reference for validation. In the false-color composite image of Sentinel-2, built-up areas appear as light purple to pinkish hues, water bodies as black, vegetation as green, and bare soil/rock as various shades of yellow. In contrast, the built-up area map employs teal blue color for built-up areas, and non-built-up areas are shown in black. In the Vembanad Lake area, AMCBI and NBI showed similar performance, except in bare paddy fields, where differences were observed (Fig. 4(b1) and (d1)). Similarly, in the outskirts of Udaipur city, NBI misclassified bare land as a built-up area (Fig. 4(d6)). Notably, all indices except AMCBI incorrectly identified high-reflective bare rock/land as built-up areas, even after thresholding (Fig. 4 (c3) to (e3)). In the case of Nainital, NDBI misclassified Nainital Lake and bare soil as built-up area (Fig. 4(e2)). However, BAEI misclassified more vegetation as built-up area than other indices in the region around Nainital Lake (Fig. 4(c2)). The visual comparison of built-up area maps derived reveals that BAEI, NBI, and NDBI exhibit a higher tendency to misclassify bare land as built-up areas, compared to AMCBI, across all six sites (Fig. 4 (b4) to (e4)). The performance of NBI is

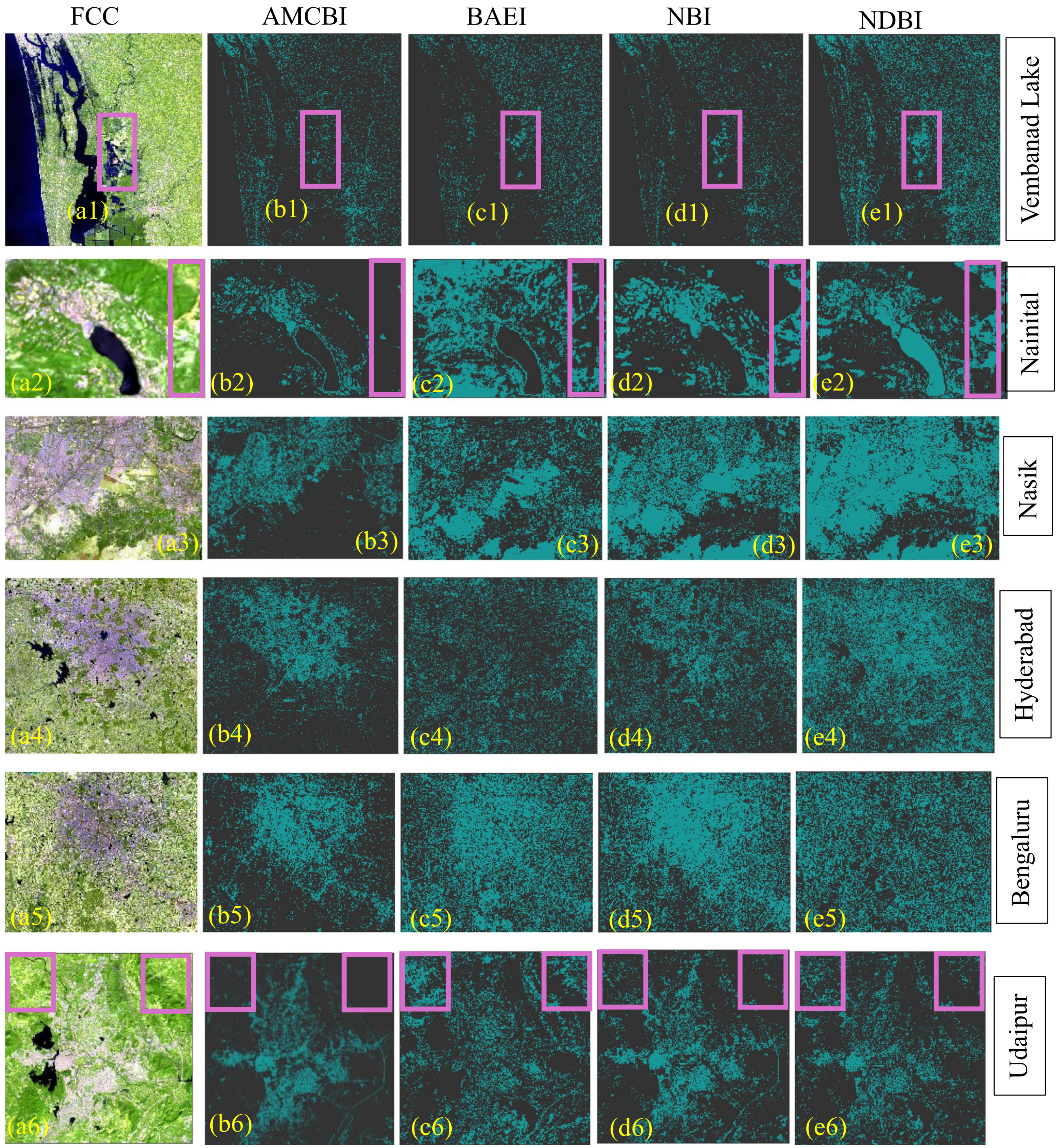

Fig. 4 Comparison of built-up area map of (**a**) False color composite of Sentinel 2 image using indices (**b**) AMBCI (**c**) BAEI (**d**) NBI (**e**) NDBI for test and validation sites (Teal blue colour represents classified built-up areas and black colour represents non built-up areas. The pink colour rectangular box highlights the areas of misclassification)

better than that of NDBI in all the sites compared in the study (Fig. 4 (d1) to (d6)).

Visual analysis of Fig. 4 reveals that the AMCBI method successfully delineates built-up areas from bare soil and vegetation without the requirement of thresholding, effectively suppressing spectrally similar bare land/soil. In contrast, the BAEI, NBI, and NDBI methods could not accurately separate built-up areas from bare soil/land, even after applying the optimal threshold values across all study areas. The incorporation of the blue band in the formulation of AMCBI effectively highlighted built-up areas due to their high reflectance. In contrast, red and NIR bands enhanced the differentiation of built-up areas from barren surfaces and vegetative regions. The visual interpretation results highlight the limitations of BAEI, NBI, and NDBI in the delineation of built-up areas from bare soil and demonstrate the superior efficiency of AMCBI in the differentiation of built-up areas from non-built-up areas. Therefore, accurate delineation of built-up areas by AMCBI prevents overestimation and provides a more precise representation of built-up area extent, unlike the other indices, which rely heavily on effective thresholding for accurate results.

Statistical evaluation of built-up index performance

Statistical analysis provides useful insights to quantify class distributions and assist in data-driven decision-making to identify patterns and trends in datasets. A frequency histogram that helps to give an overview of the dispersion of complex data is initially analyzed to evaluate the performance of the built-up mapping methods compared. Although the distribution of NBI follows the same pattern as that of NDBI (Fig. 5 (c) and (d)), the distribution of water and vegetation for NDBI is far below the zero mark compared to NBI. Further, the overlap between built-up areas and bare soil is more pronounced in the case of NBI and NDBI compared to the other indices (Fig. 5 (c) and (d)), which may lead to misclassification between built-up areas and bare soil.

However, in the case of BAEI, the close proximity in the distribution of vegetation with built-up and bare land/soil areas can be seen in Fig. 5. The effect of this can be well noted in Fig. 4 (c2) in the region where sparse vegetation co-exists with bare soil areas, where misclassification of built-up areas occurs. Both AMCBI and NDBI yield positive values for urban/built-up and bare soil areas. Moreover, NBI exhibits positive values closer to zero for water. This similarity in positive values for built-up areas and other land cover types poses a challenge for accurately delineating built-up areas, even when optimal thresholding techniques are employed. Except for AMCBI, all the indices visually categorize land cover into three types, with built-up areas and bare soil having overlapping distributions. This overlap makes the discrimination of built-up areas from other significant land cover types difficult. However, for AMCBI, a clear demarcation between the different classes (Fig. 5 (a)) could be achieved by the optimal selection and formulation of multiple spectral bands. Although AMCBI effectively distinguishes built-up areas from bare soil with minimal overlap, the constraint-based elimination process further ensures that bare soil areas are entirely eliminated, ensuring a clear distinction between the two land cover types.

To further evaluate the built-up area mapping methods, statistical tests are conducted using separability metrics such as SDI, JD, and BD values for all sites (Table 2). Table 2 shows that the JD value, which measures the separability of water from built-up areas, is consistently high (1.414) for all four methods. Furthermore, a high JD value of 1.414 indicates a good separation between vegetation and built-up areas for all methods except BAEI. However, NBI and NDBI exhibit poor separability for bare soil, with JD, SDI, and BD values below one, which indicates difficulty in the accurate identification of bare soil using these two indices.

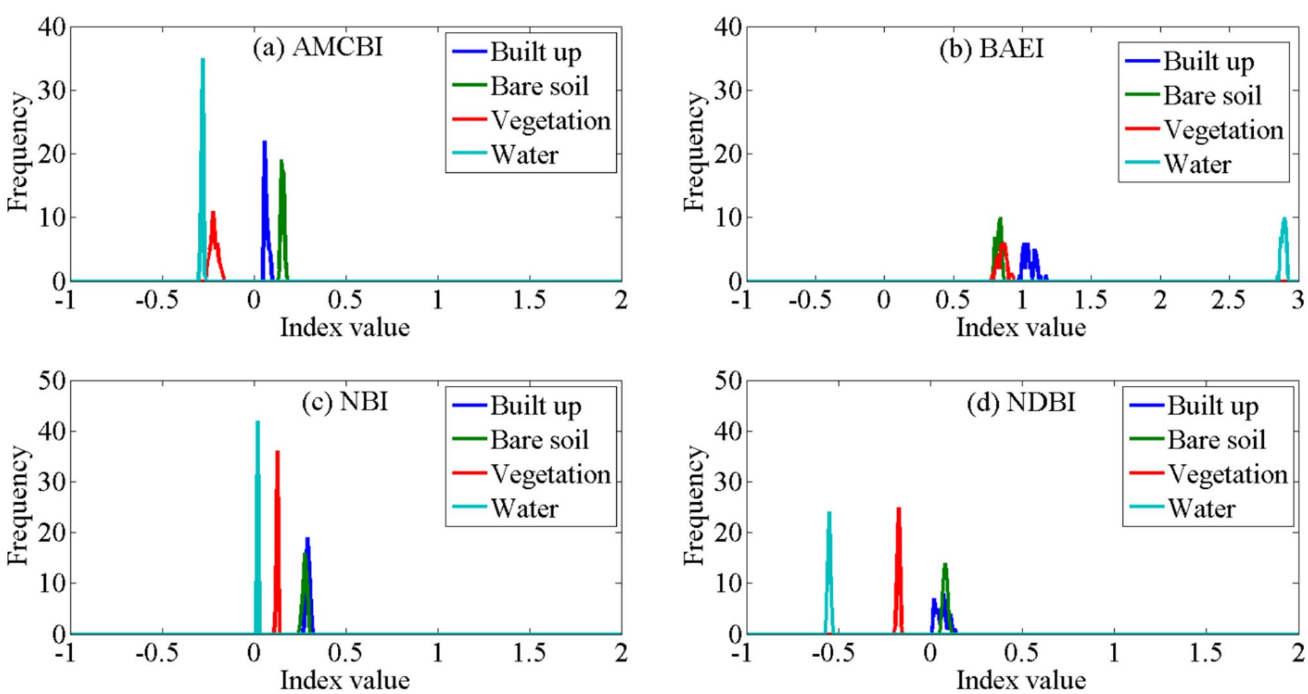

Fig. 5 Frequency histogram of (**a**) AMCBI (**b**) BAEI (**c**) NBI and (**d**) NDBI

Table 2 Separability metrics of the four indices for built up area

Indices	Bare soil			Vegetation			Water		
	SDI	JD	BD	SDI	JD	BD	SDI	JD	BD
AMCBI	5.37	1.41	14.27	9.61	1.41	40.24	27.06	1.41	315.19
BAEI	3.67	1.41	5.89	2.59	1.30	3.32	31.35	1.41	398.23
NBI	0.90	0.82	0.41	15.94	1.41	94.83	32.27	1.41	280.61
NDBI	0.29	0.71	0.45	6.35	1.41	14.56	16.86	1.41	97.37

Moreover, AMCBI, with the highest SDI (5.37) and BD (14.27) values with respect to bare soil, indicates a significantly higher ability to delineate built-up areas compared to the other three methods. The consistency of these results is further validated by the histogram plots (Fig. 5), which illustrate the variability in SDI, JD, and BD values and highlight the superior performance of AMCBI in distinguishing between land cover types. Hence, AMCBI emerges as a superior method compared to the existing ones for accurate soil mapping applications.

Assessment of classification accuracy

The assessment of the efficacy and dependability of the method employed for built-up area delineation is a critical step in accuracy assessment. Consequently, AMCBI is compared against existing built-up indices, namely BAEI, NBI, and NDBI, to evaluate its performance and determine its relative effectiveness (Table 3). A comparative analysis of built-up area mapping methods demonstrated varying levels of performance, with NBI achieving mapping accuracy between 81.2% and 93.1% and F1 scores ranging from 0.63 to 0.84. BAEI showed mapping accuracy ranging from 72.2% to 90.7% and F1 scores between 0.73 and 0.88. In contrast, the performance of NDBI has been consistently poor across all sites, with the lowest F1 score of 0.60 recorded in Bengaluru, and overall mapping accuracy and F1 scores varying from 61.3% to 91.7% and 0.60 to 0.88, respectively. The overall Accuracy (OA) values indicate that AMCBI achieved the highest accuracy of 98.7% at the test site, while BAEI, NBI, and NDBI achieved 89.7%, 91.1%, and 90.5%, respectively. Further, correlation analysis of the methods is examined to investigate any similarity between AMCBI and the other three indices (Table 4). Correlation among different built-up area indices is useful because it helps to determine whether the indices capture the same built-up area signal or respond to different aspects of urban surfaces. Although the existing built-up indices compared in this study lack correlation with each other, AMCBI shows a moderate positive correlation of 0.58 with NDBI, indicating a moderate relationship (Table 4). This reveals that, even though there is some association between them due to the incorporation of SWIR and NIR bands in AMCBI and NDBI, it is not particularly strong, and other factors may also be influencing the relationship. The low-level correlations between AMCBI and existing methods suggest a decreased likelihood of spectral overlap, resulting in reduced confusion among land-cover types, increased contrast, and more precise mapping accuracy of built-up areas. Furthermore, the better performance of AMCBI can be attributed to its enhanced separability of built-up areas from other dominant land cover types that co-exist due to the implementation of constraint-based elimination of spectrally similar bare soil areas.

Table 3 Summary of accuracy assessments of the test and validation sites

Index	Overall agreement	Quantity disagreement	Allocation disagreement	F1 score
Vembanad lake				
AMCBI	98.7	0.8	0.5	0.97
BAEI	90.7	6.9	2.4	0.88
NBI	93.1	2.9	4	0.92
NDBI	82.5	8.9	8.6	0.81
Nainital				
AMCBI	98.4	0.7	0.9	0.96
BAEI	70.2	18.6	11.2	0.68
NBI	85.1	9.4	5.5	0.84
NDBI	65.3	22.5	12.2	0.63
Nashik				
AMCBI	97.4	1.1	1.5	0.96
BAEI	84	14.8	1.2	0.83
NBI	87	6.5	6.5	0.85
NDBI	75.3	12.2	12.5	0.72
Hyderabad				
AMCBI	98.5	0.8	0.6	0.96
BAEI	72.2	16.8	1.2	0.73
NBI	81.2	7.4	11.4	0.8
NDBI	68.1	20.7	11.2	0.65
Bengaluru				
AMCBI	98.2	1.4	0.4	0.97
BAEI	85.2	8.7	6.5	0.82
NBI	90.7	8.8	0.5	0.88
NDBI	61.3	28.1	10.5	0.6
		Udaipur		
AMCBI	97	1.6	1.4	0.95
BAEI	88.2	8.3	3.5	0.85
NBI	92.3	7.2	0.5	0.9
NDBI	91.7	7.8	0.5	0.88

Table 4 Correlation matrix of built up indices

Indices	AMCBI	BAEI	NBI	NDBI
AMCBI	1.00	0.29	−0.24	0.58
BAEI	0.29	1.00	−0.05	−0.36
NBI	−0.24	−0.05	1.00	0.05
NDBI	0.58	−0.36	−0.23	1.00

Evaluation of the impact of seasonality on built-up area mapping

Seasonality plays a significant role in built-up area mapping, as changes in soil moisture and vegetation can affect mapping accuracy. Variations in soil moisture levels can modify the reflectance characteristics of bare soil and built-up areas, making it challenging to distinguish between them. The computation of built-up indices from images during dry months reduces the effectiveness of discriminating between bare soil and urban areas (Valdiviezo-N et al. 2018). Hence, understanding the impact of seasonality on built-up area mapping is crucial for developing robust and accurate land cover classification methods that can provide reliable results. In this study, three cloud-free images representing three different seasons of a subsection of the test site, namely Northeast Monsoon (09/11/2020), Winter (28/01/2021), and pre-monsoon season (04/03/2021), have been used for the seasonal analysis. The seasonal variability and its effect on the performance of four methods are assessed using the F1 score, overall agreement, quantity disagreement, and allocation disagreement, and are summarized in Table 5.

The built-up area maps derived by the four methods exhibited minimal differences in January (Fig. 6), as the prevalence of the dry and bare terrain reduced the impact of vegetation on the classification. In contrast, significant variations are seen in the built-up area maps for November and March, with notable misclassifications of bare soil as built-up areas. During November, the indices accurately eliminated high-density vegetative areas and large paddy fields but failed to account for bare soil areas with sparse vegetation, leading to an overestimation of built-up areas despite thresholding. Similarly, in March, BAEI incorrectly classified bare soil with sparse vegetation as built-up areas, while NBI and NDBI failed to distinguish even larger patches of bare soil, resulting in misclassification. The misclassification of vegetation along with built-up areas by BAEI has also been reported by Valdiviezo-N et al. (2018). The misclassifications are more prominent for all three indices except AMCBI. This discrepancy highlights the impact of seasonal variations, particularly increased vegetation cover during these months, on the accuracy and consistency of built-up area mapping. This is further verified by the accuracy assessment of the indices during different seasons (Table 5). Table 5 shows that the F1 score values exhibited minimal variability in January among the indices. However, during the cropping seasons of November and March, significant differences in both overall agreement and F1 score values are seen, primarily due to the overestimation of bare soil areas as built-up areas. The lowest F1 scores of 0.74 (NBI) and 0.72 (NDBI) in November underscore the difficulty in accurately mapping built-up areas, which requires effective separation of spectrally similar bare soil areas, further complicated by changes in reflectance due to moisture variability. Hence, it is essential to assess the robustness of the methods in the precise mapping of built-up areas that are highly season sensitive.

Table 5 Performance assessment of seasonal variability of the indices

Index	Overall agreement	Quantity disagreement	Allocation disagreement	F1 score
November				
AMCBI	96.5	1.8	1.7	0.95
BAEI	70.2	23.7	6.1	0.68
NBI	76.5	20.9	2.6	0.74
NDBI	74.5	22.9	2.6	0.72
January				
AMCBI	97.7	1.1	1.2	0.95
BAEI	93.2	2.7	4.1	0.92
NBI	91.9	5.9	2.2	0.90
NDBI	90.5	5.1	4.4	0.88
March				
AMCBI	95.9	3.6	0.5	0.97
BAEI	88.2	6.8	5.0	0.86
NBI	80.5	16.9	2.6	0.78
NDBI	76.1	12.3	11.6	0.75

Discussion

Remote sensing has enabled the development of diverse spectral indices for the swift and precise differentiation between built-up and non-built features (Estoque and Murayama 2015). However, the classification of built-up surfaces poses a significant challenge due to the similarity in the spectral reflectance of diverse and heterogeneous land cover types. Further, variations in spatial and temporal aspects and sensor heterogeneity contribute much to the inconsistent performance of various built-up area indices (Kebede et al. 2022). Currently, built-up area mapping techniques rely heavily on manual and semi-automated threshold-based methods that are location specific (Kaushik et al. 2022). Moreover, literature reveals a lack of a versatile index that consistently extracts built-up areas across different regions amidst the availability of numerous indices (Ismaeel et al., 2024).

The present study focused on the development of a methodology to delineate built-up areas without the need for thresholding and compare the performance with the commonly used built-up indices. The performance of the four

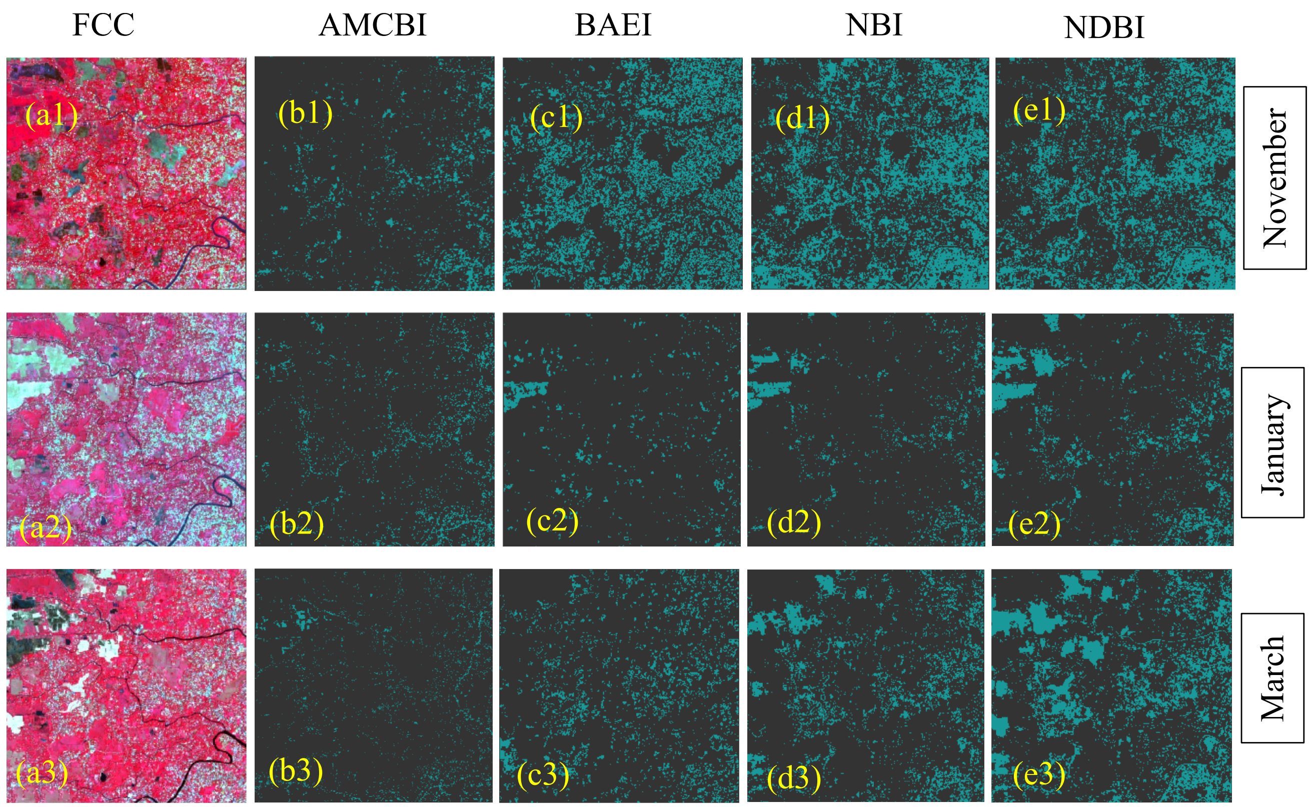

Fig. 6 Seasonality analysis of the indices in a subset of the Vembanad Lake wetland region with less density built-up areas and intense cultivation in November (with a very high percentage of vegetation), January (has a high percentage of bare soil) and March (again with more percentage of vegetation. The teal blue colour represents classified built-up areas, and the black colour represents non-built-up areas

methods has been evaluated by separability metrics, namely SDI, BD, and JD. To assess the effectiveness of built-up indices in the differentiation of built-up features, a thorough evaluation of their pixel value statistics is essential [24]. The efficacy of indices is initially evaluated using histograms of index values for four prominent land cover types (Fig. 5), which provides more insight into the separability of built-up areas from non-built-up areas. The accuracy assessment of the indices brings out the superiority of the proposed method in the delineation of built-up and urban regions (Table 3). Further, seasonality analysis demonstrated higher accuracy of BAEI over NDBI and NBI (Fig. 6), as previously reported by [24]. However, the overall comparative analysis demonstrated the superior accuracy of the proposed method over BAEI, NDBI, and NBI.

The selection of the optimal threshold plays a major role in index-based built-up area mapping methods that help to achieve robust and reliable mapping results. Nevertheless, this method necessitates substantial computational resources and manual intervention in the selection of validation points. However, the selection of the optimum threshold from overlapping values of indices corresponding to different spectrally similar land cover types is a major challenge (Kebede et al. 2022). Figure 5 indicates that most of the indices, except BAEI, could not separate the built-up areas from bare soil due to the overlap between them. Although NBI effectively differentiates vegetative areas from built-up areas, it requires further refinement to overcome spectral confusion between bare soil and built-up areas (Valdiviezo-N et al. 2018; Ismaeel et al., 2024). The NDBI method also fails to distinguish urban features from bare land due to complex spectral responses of vegetation [6] and, hence, is not effective in low vegetative areas with dry climates. However, in the case of BAEI, the overlap of built-up areas is with respect to vegetation rather than bare soil. The major shortcoming of BAEI is its failure to exclude vegetation from built-up areas, rendering it impractical to delineate built-up areas from bare land (Ismaeel et al., 2024). Although the histogram plot (Fig. 5) depicts that the built-up area and bare soil of AMCBI are clearly apart, the complete elimination of bare soil is ensured by employing the proposed constraint-based approach. Hence, by adopting the novel constraint-based methodology (AMCBI), this study could overcome the limitations of index-based thresholding and effectively address spectral overlap issues, thereby providing precise insights into urban expansion dynamics.

Even though the methodology adopted in this study not only highlights the advantages of using the newly proposed

AMCBI to monitor built-up area changes promptly using freely accessible satellite images, but also reduces uncertainty in built-up area assessment, enabling a more accurate evaluation of urban sprawl and settlement expansion over time. It can also be observed from the results that AMCBI can be effectively used in various applications such as environmental assessment to track loss of agricultural land due to urbanization, change detection studies, infrastructure, and land use planning. They also serve as inputs for assessing environmental impacts of urbanization, such as increased runoff, urban heat islands, and loss of natural land. Though the robustness of AMCBI is assessed based on its performance across different seasons, during which cloud-free data have been available, further investigation on a monthly basis can be included to thoroughly assess the index's performance at this time scale. Even small changes in vegetation growth or soil moisture can suppress or enhance the built-up signal, which makes multi-temporal comparisons more inconsistent. Another major limitation of this study is the analysis of built-up areas in snow-covered mountains, due to the limited availability of cloud-free images and the difficulty of separating clouds and snow from built-up areas at high altitude. Moreover, the scope of future work includes validating AMCBI with finer-resolution satellite data to assess its capability for built-up area extraction at a local scale, which is highly useful for urban sprawl monitoring. Further, analyses of these finer-resolution datasets for both historical and contemporary time periods help assess trends and changes in the areal extent of urbanization at the desired study scale.

Environmental application of the proposed methodology

Satellite data has been largely employed recently to differentiate built-up features from background noise, enabling the swift and precise detection and monitoring of land cover changes in urban areas. Besides built-up surface identification, remote sensing-based methodology facilitates spatial distribution analysis, although heterogeneous land cover poses serious classification challenges.

Estimation of changes in the pattern of urban expansion

An urban area is a developed region characterized by a high concentration of infrastructure, including buildings, paved surfaces, transportation systems, and other man-made structures, which can impact the quality of life and environmental health of its inhabitants. Urban sprawl, driven by changes in land use, exerts significant pressure on the ecological and socio-economic environments of surrounding areas, leading to profound impacts on natural resources, biodiversity, local population, and the economy. In the present context of rapid urban growth, consistent and accurate monitoring of urban sprawl dynamics and patterns is essential to understand its impacts and to formulate data-driven decisions to mitigate its effects and promote sustainable development. To evaluate the effectiveness of the new index in a highly urbanized environment, Bhopal city, located near the Bhoj wetland in Madhya Pradesh State, India, has been chosen. The rapidly expanding city of Bhopal is endowed with several artificial lakes, including the Upper and Lower Lakes, which are integral components of the Bhoj Wetlands. However, the tremendous growth of the city and unchecked urbanization along the periphery of the lakes have led to environmental degradation, primarily due to the influx of untreated sewage, affecting the water quality.

The results of the built-up area delineated during 2017 and 2024 for the four methods, along with the false colour composite of the Sentinel image, are presented in Fig. 7. From Fig. 7, it can be seen that, even though water could be clearly eliminated by BAEI, vegetative areas could not be eliminated even after thresholding. Although the bigger patches of bare soil devoid of vegetation are suppressed by BAEI, smaller patches of bare land with sparse vegetation are not eliminated. This resulted in a high overestimation of built-up areas compared to NBI and NDBI. The results from both NBI and NDBI are similar except in areas where mixed pixels of bare soil and built-up exist. In such areas, the performance of NBI is better than that of NDBI. However, both NBI and NDBI failed to mask out even large patches of bare soil/land, which BAEI could eliminate even after thresholding. The reference data is derived based on the manual digitization of the corresponding high-resolution Google Earth image, and the F1 score is evaluated to assess the effectiveness of the methods. The total increase in the built-up areas estimated based on Sentinel images between 2017 and 2024 is about 31.8%. The F1 score obtained for the four methods, namely AMCBI, BAEI, NBI, and NDBI, is 0.95, 0.62, 0.81, and 0.78, respectively. Though more misclassified built-up areas are visible in the case of BAEI, NBI, and NDBI, the percentage of error estimated for the built-up area is less for AMCBI. This is mainly due to the fewer interferences of the built-up surface from the bare soil area. Moreover, the performance of AMCBI is better than BAEI, NBI, and NDBI, with the highest F1 score value for AMCBI. This analysis also substantiates the superiority of AMCBI in precisely mapping urban areas, especially near highly sensitive wetland regions.

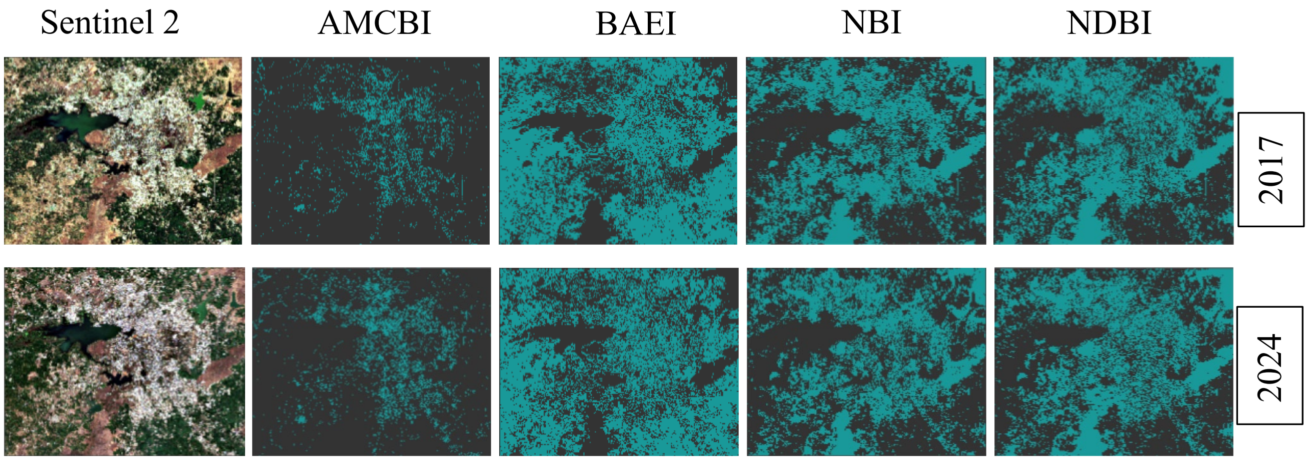

Fig. 7 Variability in a built-up area of Bhopal city in (**a**) Sentinel 2 image and using indices (**b**) AMBCI (**c**) BAEI (d) NBI (e) NDBI between 2017 and 2024 (Teal blue colour represents built-up areas classified and black colour represents non-built-up areas)

Conclusion

The present work introduces a novel constraint-based indexing approach, AMCBI, to map built-up areas utilizing Sentinel-2 imagery's blue, red, NIR, and SWIR bands. AMCBI effectively distinguishes built-up areas from non-built-up areas across diverse environmental backgrounds and climatic conditions. The integration of the constraint-based indexing approach in AMCBI has been successful in eliminating spectrally similar bare soil areas from built-up areas. The constraint-based elimination method ensured the maximum elimination of spectrally similar bare soil areas from built-up regions. The quantitative evaluation revealed that AMCBI has superior performance compared to other indices, with F1 scores of above 0.95 across different study areas chosen. In addition, the high SDI value of bare soil with respect to built-up areas for AMCBI ensured precise mapping of built-up areas. Further, the overall agreement values of above 97% and an F1 score of above 0.95 have been obtained for AMCBI across all study sites. The superiority of the AMCBI can be attributed to the ability of constraint-based indexing techniques to de-correlate built-up areas from bare land. The use of AMCBI offers great potential to achieve higher accuracy built-up area maps, which can be automated. Furthermore, the use of AMCBI eliminates the need to threshold or mask features with similar spectral characteristics as built-up areas compared to the other existing built-up indices. However, the significant limitation of this investigation is the unavailability of cloud-free monthly images for the assessment of monthly variability in the performance of the AMCBI. As precise mapping of urban areas is necessary for proper decision-making in the sectors of urban planning, resource allocation, climate change studies, and economic development, AMCBI can be employed as a robust and superior alternative to the existing built-up indices in the regions that require coordinated planning and growth management of urban areas.

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Data availability The data that support the findings of this study will be available upon request from the corresponding author.

Declarations

Competing interest The authors declare no competing interests.

References

Ali A, Nayyar ZA (2021) A modified builtup index (MBI) for automatic urban area extraction from Landsat 8 imagery. Infrared Phys Technol 116:103769

As-syakur A, Sana Adnyana IW, Arthana IW, Nuarsa IW (2012) Enhanced built-up and bareness index (EBBI) for mapping built-up and bare land in an urban area. Remote Sens 4:2957–2970

Bhatti SS, Tripathi NK (2014) Built-up area extraction using Landsat 8 Oli imagery. GISci Remote Sens 51:445–467

Bouhennache R, Bouden T, Taleb-Ahmed A, Cheddad A (2019) A new spectral index for the extraction of built-up land features from Landsat 8 satellite imagery. Geocarto Int 34:1531–1551

Bouzekri S, Lasbet AA, Lachehab A (2015) A new spectral index for extraction of built-up area using Landsat-8 data. J Indian Soc Remote Sens 43:867–873

Dabboor M, Howell S, Shokr M, Yackel J (2014) The Jeffries-Matusita distance for the case of complex Wishart distribution as a separability criterion for fully polarimetric SAR data. Int J Remote Sens 35:6859–6873

Deng C, Wu C (2012) BCI: a biophysical composition index for remote sensing of urban environments. Remote Sens Environ 127:247–259

Dixit A, Goswami A, Jain S (2019) Development and evaluation of a new "Snow Water Index (SWI)" for accurate snow cover delineation. Remote Sens 11:2774

Dozier J (1989) Spectral signature of alpine snow cover from the Landsat thematic mapper. Remote Sens Environ 28:9–22

Estoque RC, Murayama Y (2015) Classification and change detection of built-up lands from Landsat-7 ETM+ and Landsat-8 OLI/TIRS imageries: a comparative assessment of various spectral indices. Ecol Indic 56:205–217

Ettehadi OP, Kaya S, Sertel E, Alganci U (2019) Separating built-up areas from bare land in Mediterranean cities using Sentinel-2a imagery. Remote Sens 11:345

Firozjaei MK, Sedighi A, Kiavarz M, Qureshi S, Haase D, Alavipanah SK (2019) Automated built-up extraction index: a new technique for mapping surface built-up areas using LANDSAT 8 OLI imagery. Remote Sens 11:1966

Gamba P, Acqua FD (2003) Increased accuracy multiband urban classification using a neuro-fuzzy classifier. Int J Remote Sens 24:827–834

Herold M, Roberts DA, Gardner ME, Dennison PE (2004) Spectrometry for urban area remote sensing-development and analysis of a spectral library from 350 to 2400 Nm. Remote Sens Environ 91:304–319

Huete AR, Jackson RD (1987) Suitability of spectral indices for evaluating vegetation characteristics on arid rangelands. Remote Sens Environ 23:213–232

Ismaeel W A, J Satish Kumar J (2024) An approach for built-up area extraction using different indices and deep neural network (DNN) model, Infrared Physics & Technology, 142,10555

Jieli C, Manchun LI, Yongxue L, Chenglei S, Wei H (2010) Extract residential areas automatically by new built-up index. 18th International Conference in Geoinformatics on IEEE. 101109/GEOINFORMATICS20105567823.

Kaimaris D, Patias P (2016) Identification and area measurement of the built-up area with the built-up index (BUI). Int J Adv Remote Sens GIS 5:1844–1858

Kaufman YJ, Remer LA (1994) Detection of forests using mid-Ir reflectance – an application for aerosol studies. IEEE Trans Geosci Remote Sens 32:672–683

Kaushik S, Singh T, Joshi PK, Dietz AJ (2022) Automated mapping of glacial lakes using multisource remote sensing data and deep convolutional neural network. Int J Appl Earth Obs Geoinf 115:1569–8432

Kawamura M, Jayamana S, Tsujiko Y (1996) Relation between social and environmental conditions in Colombo Sri Lanka and the urban index estimated by satellite remote sensing data. Int Arch Photogramm Remote Sens 31:321–326

Kebede TA, Hailu BT, Suryabhagavan KV (2022) Evaluation of spectral built-up indices for impervious surface extraction using Sentinel-2A MSI imageries: A case of Addis Ababa city. Ethiopia Environmental Challenges 8:100568

Knight JF, Lunetta RS, Ediriwickrema J, Khorram S (2006) Regional scale land cover characterization using MODIS-NDVI 250 M multi-temporal imagery: a phenology-based approach. GiScience & Remote Sensing 43:1–23

Kumar M, Ghosh MK, Roy PK (2015) Built-up area extraction using Landsat ETM+ imagery: a comparative study of various algorithms. Int J Remote Sens 36:2646–2665

Lein JK (2006) Toward the rapid characterization of the built environment within the wildland urban interface: a soft classification strategy. GiScience & Remote Sensing 43:179–196

Liu Y, Chen J, Cheng W, Sun C, Zhao S, Pu Y (2014) Spatiotemporal dynamics of the urban sprawl in a typical urban agglomeration: a case study on Southern Jiangsu, China (1983–2007). Front Earth Sci 8:490–504

Liu Y, Meng Q, Zhang L, Wu C (2022) NDBSI: a normalized difference bare soil index for remote sensing to improve bare soil mapping accuracy in urban and rural areas. Catena 214:106265

Lo CP (2004) Testing urban theories using remote sensing. GiScience & Remote Sensing 41:95–115

Lu D, Weng Q (2004) Spectral mixture analysis of the urban landscape in Indianapolis with Landsat ETM+ imagery. Photogramm Eng Remote Sens 70:1053–1062

Lu D, Li G, Kuang W, Moran E (2013) Methods to extract impervious surface areas from satellite images. Int J Digit Earth 7:93–112

Luo X, Zhao W, Wei S, Fu Q (2015) Study on urban remote sensing classification based on improved RBF network and normalized difference indexes. Int J Signal Process Image Process Pattern Recogn 8:257–270

Lwin KK, Murayama Y (2013) Evaluation of land cover classification based on multispectral versus pansharpened Landsat ETM+ imagery. GIScience & Remote Sensing 50:458–472

Maynard CL, Lawrence RL, Nielsen GA, Decker G (2007) Modeling vegetation amount using bandwise regression and ecological site descriptions as an alternative to vegetation indices. GIScience & Remote Sensing 44:68–81

Pontius RG Jr, Millones M (2011) Death to kappa: birth of quantity disagreement and allocation disagreement for accuracy assessment. Int J Remote Sens 32(15):4407–4429

Oke TR (1982) The energetic basis of the urban heat island. Q J R Meteorol Soc 108:1–24

Pandey D, Tiwari KC (2020) Extraction of urban built-up surfaces and its subclasses using existing built-up indices with separability analysis of spectrally mixed classes in AVIRIS-NG imagery. Adv Space Res 66:1829–1845

Pereira JMC (1999) A comparative evaluation of NOAA/AVHRR vegetation indexes for burned surface detection 499 and mapping. IEEE T Geosci Remote 37:217–226

Perepechko AS, Graybill JK, Zumbrunnen C, Sharkov D (2005) Spatial database development for Russian urban areas: a new conceptual framework. GIScі Remote Sens 42:144–170

Pickard B, Gray J, Meentemeyer R (2017) Comparing quantity, allocation and configuration accuracy of multiple land change models. Land 6:52

Piyoosh AK, Ghosh SK (2018) Development of a modified bare soil and urban index for Landsat 8 satellite data. Geocarto Int 33:423–442

Powell RL, Roberts DA, Dennison PE, Hess LL (2007) Sub-pixel mapping of urban land cover using multiple endmember spectral mixture analysis: Manaus, Brazil. Remote Sens Environ 106:253–267

Pringle CM (2001) Hydrologic connectivity and the management of biological reserves: a global perspective. Ecol Appl 11:981–998

Richards JA (2013) Remote sensing digital image analysis: an introduction, 5th ed. Springer, New York

Sameen MI, Pradhan B (2016) A novel built-up spectral index developed by using multiobjective particle-swarm-optimization technique. IOP Conference Series: Earth and Environmental Science 37:012006

Small C (2003) High spatial resolution spectral mixture analysis of urban reflectance. Remote Sens Environ 88:170–186

Smitha PS, Maya K, Sudheer KP, Bindhu VM, Sreelash K, Padmalal D (2024) A novel bare soil index for enhancing the mapping of bare soil area – an indicator of urban expansion. Ecol Eng Environ Technol 25:167–180

Sun G, Chen X, Jia X, Yao Y, Wang Z (2016) Combinational buildup index (CBI) for effective built-up surface mapping in urban areas. IEEE J Sel Top Appl Earth Obs Remote Sens 9:2081–2092

Thomas N, Hendrix C, Congalton RG (2003) A comparison of urban mapping methods using high-resolution digital imagery. Photogramm Eng Remote Sens 69:963–972

Tolpekin V, Stein A (2009) Quantification of the effects of land-cover-class spectral separability on the accuracy of markov-random-field-based super resolution mapping. IEEE Trans Geosci Remote Sens 47:3283–3297

Valdiviezo-N JC, Téllez-Quiñones A, Salazar-Garibay A, López-Caloca AA (2018) Built-up index methods and their applications for urban extraction from Sentinel 2A satellite data: discussion. J Opt Soc Am A 35:35–44

Du S, Shi P, Van Rompaey A, Wen J (2015) Quantifying the impact of impervious surface location on flood peak discharge in urban areas. Nat Hazards 76(3):1457–1471

Weng Q, Lu D, Schubring J (2004) Estimation of land surface temperature-vegetation abundance relationship for urban heat island studies. Remote Sens Environ 89:467–483

Xu H (2010) Analysis of impervious surface and its impact on urban heat environment using the Normalized Difference Impervious Surface Index (NDISI). Photogramm Eng Remote Sens 76(5):557–565

Yu SS, Sun ZC, Guo HD, Zhao XW, Sun L, Wu M (2017) Monitoring and analyzing the spatial dynamics and patterns of megacities along the Maritime Silk Road. J Remote Sens 21:169–181

Yuan F, Bauer ME (2007) Comparison of impervious surface area and Normalized Difference Vegetation Index as indicators of surface urban heat island effects in Landsat imagery. Remote Sens Environ 106:375–386

Zha Y, Gao J, Ni S (2003) Use of normalized difference built-up index in automatically mapping urban areas from TM imagery. Int J Remote Sens 24:583–594

Zhao Y, Zhu Z (2022) ASI: an artificial surface index for Landsat 8 imagery. Int J Appl Earth Obs Geoinf 107:102703